ZHIYUAN CAO

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EDUCATION

Arizona State University
MS in Data Science, Analytics and Engineering — Sep 2026 - Jul 2028

Fuzhou University
Bachelor of Computer Science — Sep 2022 - Jul 2026

PUBLICATION

Zhiyuan Cao, Songyu Ke. Spatial-Temporal Representation Learning with Global-Local Alignment for Next POI Recommendation. AAAI 2026 Workshop on AI for Urban Planning

RESEARCH EXPERIENCE

Kaggle LMSYS Competition: From Battlefield to Benchmarks: Implementing LLM-as-a-Judge for Large-Scale Chatbot Preference Evaluation — May 2024 - Aug 2024
Gold Medal Winner, Ranking 7th out of 1849

- Implemented dynamic sampling with weighted random sampling to balance training batches by category proportion to decrease biased predictions due to data imbalance.
- Applied QLoRA on Gemma-2-9b-it for 4-bit quantization fine-tuning, and the PISSA initialization method to accelerate convergence, which achieved near-full-parameter fine-tuning performance on 4× A800 GPUs.
- Used the TTA augmentation strategy during testing stage, covering exchanging the input order of two models repeatedly, and performing double inference and averaged the results, which effectively eliminated 17% positional.

Spatial-Temporal Representation Learning with Global-Local Alignment for Next POI Recommendation — Aug 2025 – Nov 2025
AAAI 2026 Workshop on AI for Urban Planning

- **Local Preference Modeling**: Designed user-specific sequential hypergraphs to capture daily mobility patterns and intra-sequence correlations via hypergraph convolution and attention-based aggregation of POI embeddings to obtain local user representations.
- **Global Spatio-Temporal Modeling**: Integrated geographical graph convolution to encode spatial proximity with mobility hypergraph convolution to model inter-sequence correlations, enhanced by bi-level (POI/user) contrastive learning and gated fusion to obtain global user representations.
- **Global-Local Alignment**: Integrated distribution alignment (KL divergence on memory queue-based similarity distributions) and preference discrimination (instance-level contrastive loss) to align global and local user representations.

Undergraduate Thesis: From Global Sampling to Local Search: Combinatorial Optimization Perspective based Tabular Feature Engineering — Jan 2026 – May 2026
Outstanding Thesis Award

- **Motivation**: Decoupled symbolic feature engineering into global sampling and local search in a combinatorial space, transforming explicit feature expression generation into implicit latent-space optimization.
- **Global Sampling**: Incorporated GflowNet into the cold-start phase to sample feature expression trees proportional to downstream ML task rewards.
- **Latent Space Pretraining**: Designed a structural autoencoder with Tree-Lstm to map feature expressions into a differentiable latent space, complemented by a neural surrogate to estimate downstream performance.
- **Local Search**: Designed an iterative evolutionary framework where the neural surrogate guides multi-step gradient ascent in the continuous latent space for continuous local optimization.

OPEN-SOURCE PROJECT

FZU Bioinformatics Lab Apr 2025 – May 2025

https://github.com/zhiyuan-cao/FZU-Bioinformatics-Lab

- **Cold Start**: Built a multi-agent reinforcement learning backbone to automate high-quality feature subset selection from TCGA multi-omics data, using random forest F1-score and Laplacian score as reward signals in supervised and unsupervised modes respectively.
- **Data Collection**: Applied data augmentation via random shuffling of feature sequences to generate multiple semantically equivalent training instances from each subset, which expanded training data for subsequent model training.
- **Continuous Optimization**: Designed a sequence-to-sequence feature selection framework with BiLSTM encoder to capture contextual feature dependencies, VAE for continuous latent space smoothing, dual-head gradient ascent optimizer balancing F1 performance and redundancy, and Transformer decoder for autoregressive feature subset reconstruction.
- **Experiment**: Evaluated on TCGA multi-omics dataset (294 samples, MSI subtyping: MSS/MSI-L/MSI-H), achieved 10% higher F1 than classical feature selection methods (K-Best, mRMR, RFE, Lasso) using RF classifier, and reduced feature redundancy to 0.06% (1/100 of Lasso).

AWARDS AND HONORS

Kaggle Gold Medal (2024)
Outstanding Thesis Award (2026)

SKILLS

Languages: English (IELTS:6.5, Duolingo:115), Chinese (Native)
Programming Languages: Python / MATLAB / C++ / C / Java
Python Packages: NumPy / SciPy / sklearn / PyTorch / TensorFlow / Keras